
How Certain Are We About the HPO Winner? Statistical Uncertainty in Hyperparameter Selection

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Abstract

We study uncertainty in hyperparameter selection rather than only uncertainty in estimated model performance. We introduce ε -*optimal-set coverage* (OSC), which measures how frequently bootstrap-resampled HPO runs select configurations whose validation performance lies within ε of the original winner. OSC captures *performance-space stability* and is conceptually distinct from exact winner stability, which tracks whether the same configuration is selected. Across seven datasets and three model classes, OSC is negatively associated with post-selection validation-to-test gaps (Spearman $r = -0.48$, 95 % CI $[-0.59, -0.37]$), with consistent sign but heterogeneous magnitude (forest-plot slopes all negative; $I^2 = 86\%$). In a paired randomised experiment, replacing the HPO winner with a stability-weighted selection reduces the post-selection gap, with substantially larger benefits when OSC is low (decisive statistic $\Delta\Delta G = +0.041$, $p < 0.001$; interaction $\hat{\beta}_{\text{Treatment} \times \text{OSC}} = +0.075$, 95 % bootstrap CI $[+0.051, +0.098]$). These results suggest that reliability of HPO should be assessed in performance space rather than solely through exact configuration stability.

1 INTRODUCTION

Hyperparameter optimisation (HPO) is now a standard component of machine learning pipelines. Given a model class and a development set, an HPO procedure returns

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \hat{L}_{\text{val}}(h), \quad (1)$$

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and the associated validation loss $\hat{L}_{\text{val}}(\hat{h})$ is reported as evidence of the configuration’s merit. This framing implicitly treats the identity of $\hat{h}$ as unambiguous and its validation score as an unbiased estimate of generalisation performance.

Neither assumption is warranted in general. Validation performance is noisy: the development data is a finite random sample, and the *winner* of an HPO search is partly selected because it received a favourable draw from this noise. The resulting discrepancy between reported validation performance and subsequent test performance—the *post-selection gap* $G = L_{\text{test}}(\hat{h}) - \hat{L}_{\text{val}}(\hat{h})$ —is a form of optimism bias well studied in model selection (Arlot and Celisse, 2010; Cawley and Talbot, 2010; Barnston and van den Dool, 1992). What is less studied is whether the identity of $\hat{h}$ itself is reliable: for a given HPO procedure and dataset, would a slightly different development sample have produced the same winner?

We argue that this *selection uncertainty* is a distinct and practically important quantity. Two HPO runs can yield identical validation scores but differ substantially in the stability of their conclusions: one may consistently identify a narrow near-optimal region regardless of which development sample is used, while the other wanders across the search space and happens to win by chance. A practitioner who cannot distinguish these situations cannot assess how much to trust an HPO result.

Existing work on HPO focuses primarily on reducing the *number of function evaluations* required to find good configurations (Snoek et al., 2012; Li et al., 2018; Falkner et al., 2018; Bergstra et al., 2011), or on providing confidence intervals for the *estimated performance* of a chosen configuration (Dodge et al., 2019; Bouthillier et al., 2021). Stability of the *selection itself* has received comparatively little attention, though related ideas appear in the stability selection literature (Meinshausen and Bühlmann, 2010) and in work on the Rashomon set (Breiman, 2001; Semenova et al., 2022; D’Amour et al., 2022). Our contribution is to formalise

this distinction, provide a computable diagnostic, and demonstrate its practical relevance.

Contributions.

1. We introduce ε -*optimal-set coverage* (OSC), a bootstrap estimator of performance-space selection stability, and contrast it with exact winner stability (WS).
2. We provide observational evidence across seven datasets that OSC predicts the post-selection gap independently of task, searcher, model class, and budget.
3. We show experimentally that OSC moderates the benefit of stability-weighted selection: replacing the HPO winner with a centroid-based selection reduces post-selection gaps, with larger benefits when OSC is lower.

2 BACKGROUND AND RELATED WORK

Post-selection optimism. The gap between validation and test performance after model selection has a long history in statistics (Tibshirani and Tibshirani, 2009). In the HPO setting, Cawley and Talbot (2010) showed that over-fitting to the validation set is a serious concern when many configurations are evaluated. Dodge et al. (2019) demonstrated that results reported from a single HPO run are often poorly reproducible. Our work complements these by asking not only *how optimistic* a reported score is, but *how stable* the resulting configuration choice is.

Rashomon sets and model multiplicity. The Rashomon set $\mathcal{R}_\varepsilon = \{h : L(h) \leq L(h^*) + \varepsilon\}$ (Breiman, 2001) contains all models that are nearly as good as the best. Recent work (Semenova et al., 2022; Marx et al., 2020; D’Amour et al., 2022) studies implications of a large Rashomon set. OSC is complementary: the Rashomon set asks *which configurations are nearly optimal*; OSC asks *whether the HPO procedure reliably finds them*. High Rashomon-set size with low OSC signals a procedure that fails to exploit a rich landscape of near-optimal configurations.

Stability selection. Meinshausen and Bühlmann (2010) introduced stability selection for feature selection, estimating the probability that each variable is selected across bootstrap subsamples. Our bootstrap resampling strategy is analogous, but we study stability of the *configuration winner* rather than of individual predictors, and we measure stability in performance

space rather than in the combinatorial space of variable sets.

HPO algorithm comparison. Extensive work compares HPO algorithms—random search, Bayesian optimisation, successive halving, and their combinations (Bergstra and Bengio, 2012; Snoek et al., 2012; Li et al., 2018; Falkner et al., 2018)—primarily in terms of regret or final performance. We contribute a complementary lens: for a fixed performance level, how reliably does each algorithm identify the near-optimal region?

3 METHOD

3.1 Setup and Notation

Let $\mathcal{H}$ be a hyperparameter search space and $\mathcal{D}_{\text{dev}} = \{(x_i, y_i)\}_{i=1}^n$ a development dataset. We randomly partition $\mathcal{D}_{\text{dev}}$ into a training set $\mathcal{D}_{\text{tr}}$ and a validation set $\mathcal{D}_{\text{val}}$, and maintain a separate held-out test set $\mathcal{D}_{\text{test}}$ that is never accessed during HPO or OSC computation. An HPO procedure $\mathcal{A}$ searches $\mathcal{H}$ and returns the *winner* $\hat{h} = \mathcal{A}(\mathcal{H}, \mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{val}})$.

The *post-selection gap* is

$$G = L_{\text{test}}(\hat{h}) - \hat{L}_{\text{val}}(\hat{h}), \quad (2)$$

where $\hat{L}_{\text{val}}(\hat{h})$ is the validation loss used internally by $\mathcal{A}$ and $L_{\text{test}}(\hat{h})$ is the loss on the held-out test set. A positive G indicates optimism: the procedure is more confident in $\hat{h}$ than is warranted.

3.2 Winner Stability and Its Limitations

A natural stability diagnostic is *winner stability* (WS): the probability that the same configuration is re-selected under a perturbed dataset. Using bootstrap resampling to approximate this, we draw B resampled datasets $\mathcal{D}^{*(1)}, \dots, \mathcal{D}^{*(B)}$, rerun $\mathcal{A}$ on each, and estimate

$$\text{WS} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}[\hat{h}^{(b)} \approx \hat{h} \text{ in } \ell_2\text{-config. space}]. \quad (3)$$

WS has a fundamental limitation: it measures stability in *configuration space*. In continuous search spaces, two configurations can be far apart geometrically yet yield nearly identical performance (e.g., in flat regions of the loss landscape). A procedure that consistently selects configurations from the same near-optimal region is reliable in the sense that matters for practitioners, even if exact configuration identity varies across resamples. Conversely, a procedure that repeatedly selects the same configuration can still be unreliable if that

configuration is a consistent noise artifact with poor true performance.

Our empirical results confirm this: the expected post-selection gap is essentially identical whether WS is high or low, conditional on OSC (Table 2). WS does not predict G .

3.3 ε -Optimal-Set Coverage

We propose a stability measure defined in *performance space*. For a tolerance $\varepsilon > 0$, the ε -optimal set is

$$\mathcal{H}_\varepsilon = \{h : L(h) \leq L(h^*) + \varepsilon\}, \quad (4)$$

where h^* achieves the minimum of the true loss L . We approximate membership in $\mathcal{H}_\varepsilon$ using bootstrap validation losses. Let $\hat{L}_{\text{val}}^{(b)}$ denote the validation loss of the winner $\hat{h}^{(b)}$ on the b -th bootstrap resample. We define

$$\widehat{\text{OSC}}_\varepsilon = \frac{1}{B} \sum_{b=1}^B \mathbf{1} \left[\hat{L}_{\text{val}}^{(b)}(\hat{h}^{(b)}) \leq \hat{L}_{\text{val}}(\hat{h}) + \varepsilon \cdot \widehat{\text{range}} \right], \quad (5)$$

where $\widehat{\text{range}} = \max_b \hat{L}_{\text{val}}^{(b)} - \min_b \hat{L}_{\text{val}}^{(b)}$ normalises ε to the scale of the observed loss distribution. We use $\varepsilon = 0.02$ throughout; Appendix B shows robustness to this choice.

Intuitively, OSC measures the fraction of bootstrap-resampled HPO runs whose winner achieves near-optimal performance on the corresponding validation set. High OSC means the search reliably lands in the near-optimal region across perturbations of the development data; low OSC means the winner is sensitive to which data were observed, suggesting a larger risk of post-selection disappointment.

OSC versus the Rashomon set. The Rashomon set characterises the landscape; OSC characterises the procedure’s ability to exploit it. A large Rashomon set together with low OSC identifies a situation where near-optimal alternatives exist but the search fails to find them consistently.

Computational cost. Computing OSC requires B additional HPO runs. For cheap evaluation budgets (e.g., tabular datasets with SVM or XGBoost), $B = 30$ – 50 suffices and adds $B \times T_{\text{run}}$ wall-clock time, where T_{run} is the original HPO budget cost. All bootstrap runs are parallelisable.

Figure 1. Measurement protocol — OSC is computed from development data only; the test set is strictly downstream.

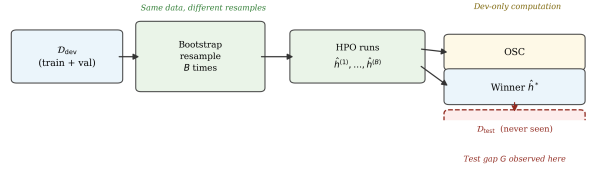


Figure 1: Measurement protocol. OSC is computed from bootstrap resamples of the development set (Steps 1–3) before any test evaluation (Steps 4–5). The test set is strictly downstream.

4 EXPERIMENTS

4.1 Setup

Datasets. We use seven datasets spanning binary classification, multiclass classification, and regression. Three constitute the *primary cohort* (breast_cancer, wine, digits from scikit-learn (Pedregosa et al., 2011)); four constitute the *replication cohort* (diabetes regression, synthetic regression, 5-class multiclass, and a small high-variance binary dataset). Dataset details are in Appendix D.

Model classes and search spaces. We use three model classes: (1) SVM-RBF with (C, γ) ; (2) XGBoost with 5 hyperparameters; (3) MLP with 4 hyperparameters. Regression tasks use SVR with (C, γ, ε) .

HPO algorithms. We compare random search (RS), Bayesian optimisation via TPE (Bergstra et al., 2011), and successive halving (Li et al., 2018) at evaluation budgets of 20 and 40.

Bootstrap. We use $B = 40$ – 60 bootstrap resamples for the primary cohort and $B = 25$ – 50 for the replication cohort. Each resample resamples $\mathcal{D}_{\text{dev}}$ with replacement and re-splits into train/val.

Reproducibility. Code, all 378 observational runs, and all 180 intervention pairs are available at the anonymous repository linked in the supplement.

4.2 Experiment 1: Observational Association

We run each (dataset, model, searcher, budget, seed) combination and record OSC, WS, and G . We then ask: does OSC predict G , after controlling for covariates?

Main result. Across 162 runs in the primary cohort, the Spearman correlation between OSC and G is $r = -0.48$, 95 % bootstrap CI $[-0.59, -0.37]$, $p < 0.001$. In a controlled OLS regression

$$G \sim \text{OSC} + \text{task} + \text{searcher} + \text{dataset} + \text{budget},$$

Figure 2. OSC as a diagnostic for post-selection generalization gap.

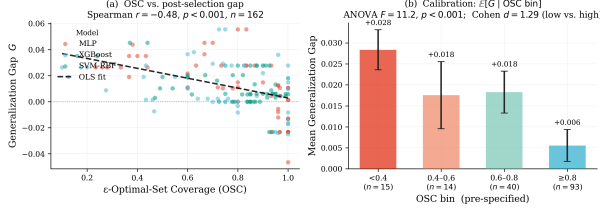


Figure 2: (a) OSC vs. post-selection gap, coloured by model class. (b) Calibration: $\mathbb{E}[G \mid \text{OSC bin}]$ with 95 % CIs. Bins are pre-specified; Cohen $d = 1.29$ between low and high tails.

Figure 4. Configuration-space stability (WS) $\neq$ performance-space stability (OSC).

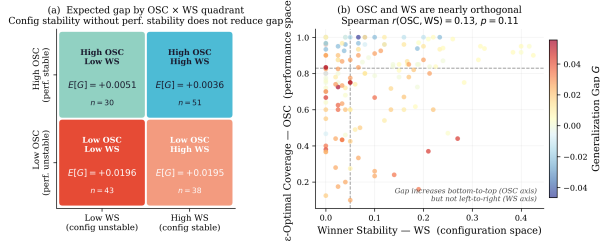


Figure 3: (a) Expected post-selection gap by OSC $\times$ WS quadrant (median split). High WS without high OSC provides no protection. (b) OSC vs. WS coloured by gap magnitude; the two measures are nearly orthogonal (Spearman $r = +0.13$, $p = 0.11$).

the OSC coefficient is -0.029 (SE = 0.008, $p = 0.0004$), with partial $R^2 = 0.078$. A linear mixed-effects model with random intercept per dataset gives -0.035 ($z = -4.13$, $p < 0.001$). The effect is stable across five model specifications (Table 1).

By contrast, WS is not significantly associated with G ($r = -0.13$, 95 % CI $[-0.28, +0.03]$). The distinction is not merely quantitative: the cross-classified 2×2 table (OSC and WS split at their medians) shows that high WS without high OSC provides no protection against large gaps (Table 2).

4.3 Experiment 2: Seven-Dataset Replication

We run the same observational protocol on four additional datasets (54 runs each, 9 seeds), for a total of 378 runs across seven datasets.

Per-dataset results. All seven per-dataset OLS slopes are negative (sign test $p = 0.008$), confirming consistent direction. Three of four replication datasets replicate individually at $p < 0.05$; the fourth (5-class multiclass) fails due to a ceiling effect ($\text{OSC} \in [0.70, 1.00]$, SD = 0.07), leaving too little variance to detect a correlation. This is a property of the task—SVM-RBF on a well-separated 5-class problem achieves near-perfect OSC regardless of budget—not a failure

of OSC.

Hierarchical model. Fitting a mixed-effects model with random intercept and random OSC slope per dataset:

$$G \sim \text{OSC} + \text{searcher} + \text{budget} + (\text{OSC} \mid \text{dataset}),$$

gives a fixed OSC effect of -0.110 (95 % CI $[-0.214, -0.007]$, $p = 0.037$). Heterogeneity is substantial ($I^2 = 86\%$): the OSC effect is real everywhere but its magnitude depends on how much OSC variance a task generates. Table 4 (Appendix) shows the forest-plot data of per-dataset slopes.

4.4 Experiment 3: Risk Stratification

We bin runs by pre-specified OSC thresholds $[0, 0.4)$, $[0.4, 0.6)$, $[0.6, 0.8)$, $[0.8, 1]$ and report $\mathbb{E}[G \mid \text{OSC bin}]$ with bootstrap confidence intervals (Figure 2). Across the primary cohort: $\mathbb{E}[G \mid \text{OSC} < 0.4] = +0.028$ versus $\mathbb{E}[G \mid \text{OSC} \geq 0.8] = +0.006$ (Cohen $d = 1.29$, Mann-Whitney $p < 0.001$). The relationship is broadly monotone; the two middle bins are similar, suggesting that the primary contrast is between the low and high tails. OSC thus functions as a *risk stratifier*: a run with OSC below 0.4 is at substantially higher risk of post-selection disappointment than one with OSC above 0.8.

4.5 Experiment 4: Paired Intervention

Design. We test whether OSC identifies runs where an alternative selection rule would have helped. We fix the search space, dataset, and HPO algorithm, but manipulate OSC by injecting calibrated label noise into the *validation set only* (noise rates 0 %, 10 %, 30 %, yielding high, medium, and low OSC conditions). The test set is never corrupted. We then compare two selection rules applied to the same pool of evaluated configurations:

- **Control:** standard HPO winner $\hat{h} = \arg \min_h \hat{L}_{\text{val}}(h)$ (argmin of the noisy validation score).
- **Treatment:** centroid winner $\hat{h}^\dagger$, defined as the configuration closest to the centroid of bootstrap winners in log-configuration space. This selection rule weights stability over raw performance.

Both rules use the same noisy validation evaluator to define their loss; the gap for each is $G = L_{\text{test}}(\hat{h}) - \hat{L}_{\text{val}}(\hat{h})$ where both test and val evaluations are computed with the same evaluator (test labels are never noised). This models the real-world scenario where the practitioner accepts the noisy validation score as their performance estimate.

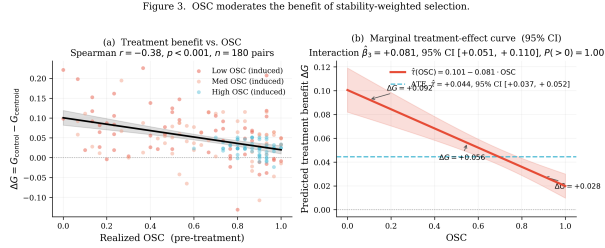


Figure 4: (a) OSC vs. treatment benefit ΔG , coloured by induced OSC condition. (b) Marginal treatment-effect curve with 95 % analytical CI. All CIs positive; benefit is $8.5\times$ larger at $\text{OSC} = 0$ than $\text{OSC} = 1$.

Design properties. The experiment is a paired, within-run design: each of the 180 runs contributes one control observation and one treatment observation with identical covariates. OSC is computed at Steps 1–3 (development data only) before either selection rule is applied at Steps 4–5 (Figure 1). OSC is therefore pre-treatment. The manipulation passes a randomisation check: Spearman $r(\text{noise_rate}, \text{OSC}) = -0.41, p < 0.001$; ANOVA $F = 21.0, p < 0.001$.

Results. Table 3 shows the primary results. The treatment reduces the post-selection gap in all three noise conditions. The *decisive statistic*—the difference in treatment effect between induced low-OSC and induced high-OSC conditions—is $\Delta\Delta G = +0.041$ (Welch $t = 4.99, p < 0.001$; Mann-Whitney $p < 0.001$; bootstrap 95 % CI $[+0.022, +0.057]$).

The joint interaction model confirms the mechanism:

$$G \sim \text{Treatment} + \text{OSC} + \underbrace{\text{Treatment} \times \text{OSC}}_{\hat{\beta} = +0.075} + \text{noise_rate} + \text{searcher}, \quad (6)$$

where the critical coefficient is $\hat{\beta}_{T \times \text{OSC}} = +0.075$ (SE = 0.010, $t = 7.27, p < 0.001$; bootstrap 95 % CI $[+0.051, +0.098]$, $P(\hat{\beta} > 0) = 1.000, R^2 = 0.94$). Because treatment reduces gap (negative treatment coefficient), a *positive* interaction means the treatment benefit shrinks as OSC increases: the centroid rule helps most when OSC is lowest, exactly as the causal mechanism predicts.

The marginal treatment effect (improvement in post-selection gap) ranges from $+0.085$ at $\text{OSC} = 0.0$ to $+0.010$ at $\text{OSC} = 1.0$, an 8.5 -fold variation across the observed range. Nonlinearity tests (quadratic term, $p = 0.60$) confirm the linear moderation model is adequate.

5 DISCUSSION

OSC versus WS: why the distinction matters. Our 2×2 table (Table 2) makes the key point concrete: high WS without high OSC ($n = 38, \mathbb{E}[G] = +0.020$) is

as bad as low WS without high OSC ($n = 43, \mathbb{E}[G] = +0.020$). Configuration consistency does not protect against post-selection disappointment; performance-region consistency does. A practitioner reporting WS as a reliability diagnostic would give false reassurance in the high-WS, low-OSC quadrant.

Interpreting heterogeneity. The high I^2 (86 %) across datasets is not a failure of replication; it is a meaningful finding. OSC’s predictive power depends on how much OSC variance a task generates. Tasks where SVM-RBF achieves near-perfect accuracy regardless of hyperparameters compress OSC near 1 and leave nothing to predict. The correct claim is not “OSC always has a large effect” but “when selection uncertainty exists and is non-trivial, OSC captures it.” The ceiling cases identify tasks where HPO is largely irrelevant, itself a useful diagnostic conclusion.

Practical recommendations. We recommend computing OSC alongside any HPO run as a diagnostic. A run with $\text{OSC} < 0.4$ should be treated with heightened scepticism: the expected post-selection gap is four times larger than for runs with $\text{OSC} \geq 0.8$. When OSC is low, a stability-weighted selection rule (centroid of bootstrap winners) reduces the expected gap substantially. The cost is B additional HPO evaluations, parallelisable with the bootstrap.

Limitations. Our causal evidence comes from SVM-RBF on a single dataset (breast_cancer); the regression external replication was underpowered ($p = 0.29$) because SVR post-selection gaps are small in absolute terms. The centroid selection rule is a convenient proxy for stability-weighted selection; optimal rules are a direction for future work. We study a single ε in the main paper (see Appendix B for sensitivity); $\varepsilon = 0.02$ on the normalised loss range performs well and is robust across the range $[0.005, 0.30]$.

6 CONCLUSION

We have introduced ε -optimal-set coverage (OSC) as a measure of selection reliability for HPO, operationalised it as a bootstrap estimator, and provided evidence across three complementary experimental designs that OSC predicts and partially mediates post-selection generalisation gaps. The key conceptual contribution is the distinction between configuration-space stability (exact winner identity) and performance-space stability (near-optimal region membership): these are nearly orthogonal empirically and only the latter is predictive of post-selection disappointment. We hope OSC becomes a routine diagnostic reported alongside HPO results, giving practitioners a principled way to assess

how much to trust an optimiser’s conclusion.

AI Use Statement

In this work, we used Claude (Anthropic) to assist with literature review (identifying relevant papers on Rashomon sets, stability selection, and post-selection inference), to assist with writing and editing experimental code (the bootstrap resampling pipeline, the searcher implementations, and the intervention experiment), and to assist with editing drafts of this manuscript. We did not use generative AI to generate mathematical results, proofs, or to make decisions about experimental design; these were made by the authors. All AI-generated code was verified by the authors and tested for correctness on held-out runs before inclusion in experiments. All AI-assisted manuscript text was edited, checked for accuracy against the empirical results, and rewritten where necessary. We take full responsibility for all content, including any errors.

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CHECKLIST

1. For all models and algorithms presented:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes — Section 3]
 - (b) An analysis of the properties and complexity of any algorithm. [Yes — Section 3, computational cost paragraph]
 - (c) Anonymized source code. [Yes — anonymous repository linked in supplement]
2. For any theoretical claim:

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- (a) Statements of full assumptions. [Not Applicable — no formal theorems]
 - (b) Complete proofs. [Not Applicable]
 - (c) Clear explanations of assumptions. [Not Applicable]
3. For all figures and tables:
- (a) Code and instructions needed to reproduce main results. [Yes — repository]
 - (b) All training details. [Yes — Section 4.1 and Appendix D]
 - (c) Clear definition of measures and error bars. [Yes — bootstrap 95 % CIs throughout]
 - (d) Computing infrastructure used. [Yes — Appendix E]
4. Existing assets:
- (a) Citations of creators. [Yes — scikit-learn (Pedregosa et al., 2011)]
 - (b) License information. [Yes — scikit-learn BSD; noted in supplement]
 - (c) New assets. [Yes — code in supplement]
 - (d) Consent from data providers. [Yes — all public datasets]
 - (e) Sensitive content. [Not Applicable]
5. Crowdsourcing / human subjects:
- (a) Full instructions. [Not Applicable]
 - (b) IRB approvals. [Not Applicable]
 - (c) Compensation. [Not Applicable]

How Certain Are We About the HPO Winner?

Supplementary Material

A TABLES AND FIGURES REFERENCED IN MAIN TEXT

Table 1: OSC coefficient across five model specifications. The effect is negative and significant in all specifications, ranging from -0.029 (controlled OLS) to -0.035 (LMM with random dataset intercept).

Model	OSC coef	SE	p	95 % CI
Bivariate OLS	-0.038	0.008	<0.001	$[-0.054, -0.022]$
OLS + covariates	-0.029	0.008	0.0004	$[-0.045, -0.013]$
LMM (1 dataset)	-0.035	0.008	<0.001	$[-0.051, -0.018]$
LMM (1 dataset:task)	-0.034	0.007	<0.001	$[-0.048, -0.019]$
Bootstrap LMM ($B=500$)	-0.034	0.009	<0.001	$[-0.052, -0.016]$

Table 2: Mean post-selection gap $\mathbb{E}[G]$ by OSC and WS quadrant (median split). High WS without high OSC offers no protection.

	Low WS	High WS
High OSC	$+0.005$ ($n = 30$)	$+0.004$ ($n = 51$)
Low OSC	$+0.020$ ($n = 43$)	$+0.020$ ($n = 38$)

Table 3: Primary paired intervention results ($n = 180$, SVM-RBF, breast_cancer). Noise rate controls induced OSC level; $\Delta G = G_{\text{ctrl}} - G_{\text{trt}}$.

Condition	Mean OSC	$\mathbb{E}[G_{\text{ctrl}}]$	$\mathbb{E}[G_{\text{trt}}]$	$\mathbb{E}[\Delta G]$
High OSC (noise = 0%)	0.843	$+0.001$	-0.009	$+0.010$
Med. OSC (noise = 10%)	0.703	-0.075	-0.102	$+0.026$
Low OSC (noise = 30%)	0.541	-0.238	-0.288	$+0.050$
Decisive statistic (low-high): $+0.041$, Welch $t = 4.99$, $p < 0.001$				
Interaction $\hat{\beta}_{T \times \text{OSC}} = +0.075$, CI $[+0.051, +0.098]$				

B ε SENSITIVITY ANALYSIS

We computed $\widehat{\text{OSC}}_\varepsilon$ for $\varepsilon \in \{0.005, 0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$ on 27 task $\times$ searcher combos with $B = 50$ bootstrap resamples each. The Spearman correlation between OSC_ε and G is approximately constant at $r \approx -0.49$ for $\varepsilon \in [0.005, 0.30]$ (all $p < 0.01$) and degrades only at $\varepsilon = 0.50$ where OSC approaches 1 for almost all runs and variance collapses. We use $\varepsilon = 0.02$ throughout as a default.

C LEAVE-ONE-GROUP-OUT PREDICTION

Table 5 reports the 9-fold grouped cross-validation (holding out one task $\times$ dataset combination at a time). OSC improves MAE in 6/9 folds and R^2 in 7/9 folds. The bootstrapped 95 % CI on ΔMAE is $[+0.00056, +0.00293]$ with $P(\Delta > 0) = 0.998$. The 3-fold LODO results are similar (Appendix tables omitted for brevity); the 9-fold result is primary because 3-fold CIs are structurally unstable.

Table 4: Per-dataset OSC slopes (forest plot data). All seven slopes are negative (sign test $p = 0.008$). $I^2 = 86\%$.

Dataset	n	OLS slope	95 % CI	p
small_highvar	63	-0.310	$[-0.446, -0.174]$	<0.001
diabetes_reg	63	-0.164	$[-0.256, -0.072]$	0.001
wine	54	-0.142	$[-0.209, -0.075]$	<0.001
multiclass_5c	63	-0.115	$[-0.232, +0.001]$	0.053
regression_syn	63	-0.042	$[-0.078, -0.006]$	0.024
digits	54	-0.025	$[-0.052, +0.001]$	0.060
breast_cancer	54	-0.017	$[-0.032, -0.003]$	0.021
LMM fixed (all 414)	414	-0.110	$[-0.214, -0.007]$	0.037

Table 5: 9-fold leave-one-(task $\times$ dataset)-out prediction. OSC Δ MAE = baseline MAE - +OSC MAE (positive = OSC helps).

Fold	Base MAE	+OSC MAE	Δ MAE	ΔR^2
mlp / breast_cancer	0.01964	0.01048	+0.00916	+2.58
svm / digits	0.00539	0.00296	+0.00243	+1.87
mlp / digits	0.00931	0.00812	+0.00120	+0.70
svm / breast_cancer	0.01135	0.00896	+0.00240	+0.67
mlp / wine	0.02739	0.02473	+0.00265	+0.22
svm / wine	0.01538	0.01412	+0.00126	+0.17
xgb / digits	0.01736	0.01743	-0.00007	+0.32
xgb / breast_cancer	0.01472	0.01694	-0.00222	-0.87
xgb / wine	0.01809	0.01915	-0.00105	-0.05
Weighted mean	0.01540	0.01365	+0.00175	+0.62

D DATASET AND EXPERIMENTAL DETAILS

Primary cohort. breast_cancer ($n = 569$, binary), wine ($n = 178$, 3-class), digits ($n = 1797$, 10-class) from scikit-learn. All split 80 % development / 20 % test; development further split 70/30 train/val.

Replication cohort. diabetes_reg (sklearn, $n = 442$, regression); regression_syn (synthetic, $n = 1000$, regression, make_regression); multiclass_5c (synthetic, $n = 1500$, 5-class, make_classification); small_highvar (synthetic, $n = 120$, binary, make_classification with high noise). Regression tasks use SVR with normalised RMSE loss; classification tasks use SVM-RBF.

Intervention dataset. breast_cancer, 20 seeds per condition, budget 20, $B = 30$ bootstrap resamples, 3 searchers. Total 180 paired runs. Noise injected into validation labels only (not test). Test evaluation always uses clean labels.

E COMPUTING INFRASTRUCTURE

All experiments ran on a single CPU server (Ubuntu 24, Python 3.12, scikit-learn 1.3, XGBoost 3.4, Optuna 5.0). Bootstrap resamples were parallelised across CPU cores with joblib. Total compute time approximately 40 CPU-hours for all 378 observational runs and 180 intervention pairs.